

HOW A QUASILIQUID INTERFACE PROMOTES RESILIENCE OF ICE CRYSTAL FACETING

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Abstract. We explore the hypothesis that a key factor in determining the dynamics and morphology of faceted ice-vapor surfaces is the quasi-liquid layer that forms at this interface at temperatures above 240 K. We do so by modeling the ice surface as reaction-diffusion system in which the time scales of quasi-liquid freezing and melting, horizontal diffusion, and exchanges with the vapor phase are made explicit. Model parameterizations are chosen to represent vapor fields around growing and ablating ice crystals. Our results support a diffusive slowdown mechanism in which regions of the ice-vapor interface that are exposed to higher supersaturations spontaneously undergo structural modifications spontaneously that render them less efficient at retaining surface admolecules. The result manifests as a limit cycle in the thickness of the quasiliquid layer, a traveling wave solution for the underlying ice, and a constant rate of growth across a crystal facet overall. The same model, when subsaturated conditions are imposed, is found to promote faceted sublimation.

Key words. ice, surface, PDE

MSC codes. 35K57, 37M05, 70K05

1. Introduction. Ice crystals that make up cirrus clouds modulate Earth’s climate by scattering sunlight before it reaches Earth’s surface. The extent to which a given cirrus cloud does so depends not only on the number density of ice crystals in the cloud, but also on the morphologies of the crystals. Although hexagonal prisms dominate, natural cirrus clouds exhibit a great deal of variation, including long and thin hexagonal needles, short and wide plates, indentations, hollowing, and even dendritic forms [1, 2]. Moreover, the faceted surfaces of ice crystals can be rough on a scale that matters to the wavelength of sunlight [3]. Because of this variability, integrating ice crystal morphologies into climate models remains a formidable task, and contributes significantly to uncertainties in climate change prediction.

All this points to a need for better models of ice surface morphology and dynamics. Atomistic-scale studies have contributed greatly to this effort (see, e.g., reference [4]). “Mesoscale” models, defined here as models spanning distances of micrometers to hundreds of micrometers, have also been developed, of which Harrington and Pokrifka [5] provide an excellent review. Two approaches that combine insight on both of these scales that are particularly relevant to the current work are Classical Nucleation Theory and Continuum Theory.

In Classical Nucleation Theory (CNT), the atomistic view of the process of crystal growth can be summarized as follows: when a gas-phase water molecule encounters a crystalline ice surface, it initially becomes attached to that surface as an “admolecule.” Not yet part of the crystal’s lattice, the admolecule diffuses across the surface until it meets one of two fates: either it fills an unoccupied position in the crystal lattice, or else it returns to the gas phase. New layer formation is governed by a 2D nucleation process in which the step free energy plays a central role [6].

CNT has long framed how we think about ice crystal growth from the vapor phase, but it suffers from some significant drawbacks. First, it offers little guidance

when it comes to describing the behavior of crystals placed in a vapor field that is inhomogeneous on a mesoscopic scale. To be specific, when a growing faceted ice crystal is situated in a supersaturated vapor field, the expectation is that the surrounding water vapor concentration will be drawn down in such a way that protruding corners will experience persistently higher vapor pressures than facet centers. This pattern of higher concentrations immediately above a growing crystal surface was first noted by Berg [7], in the context of crystal growth from solutions, and will therefore be termed “Bergian” (see discussion below). CNT predicts that Bergian growth conditions would result in faster growth at those protruding corners, leading ultimately to dendritic forms (similar to snowflakes). However, under typical cirrus cloud conditions, it is known that hexagonal prisms *resist* that tendency, despite those inhomogeneities. Classical nucleation theory has no satisfactory explanation for the persistence of hexagonal prismatic habit in the presence of inhomogeneities in the overlying vapor. (However, see reference [8]).

A second shortcoming of CNT concerns the presence of a quasi-liquid layer on the surface. When the temperature of ice rises above 240 K, both experiment and theoretical studies have shown that the ice/air interface is covered by a quasi-liquid layer (QLL). Moreover, molecular dynamics studies have shown that gas-phase water molecules striking the QLL surface are thermalized quickly – on a picosecond time scale [9]. While the presence of a QLL does not preclude crystalline layering (recent experimental work has shown that steps and ledges may form *underneath* the QLL, at the interface between the QLL and the underlying ice [10]), CNT is ill-equipped to describe such processes.

A quasi-liquid-continuum model, here referred to as QLC-1, was presented by some of the authors in Neshyba et al [11] (henceforth N2016) as a way to address these shortcomings. The main quantities and processes are indicated in Fig. 1. Dynamical variables N_{tot} and N_{QLL} (both functions of time and position on the ice surface) represent the total thickness of the ice surface and its quasi-liquid part, respectively. The two labeled points on the surface represent limiting cases. “ μ surface I” is the thinnest and least volatile possible microsurface, while “ μ surface II” is the thickest and most volatile microsurface. Between these two we imagine a continuum of possibilities, intermediate in thickness and volatility. The areal extent of each of microsurface is given by dx , the discretization width that will be used in the spatially finite representation of the surface. The evolution of the surface over time is governed by a pair of reaction-diffusion equations representing the processes indicated by arrows in Fig. 1. These take into account exchange of QLL molecules with the vapor phase (deposition and sublimation), diffusion of QLL molecules across the ice surface, and conversion of QLL molecules to and from the underlying ice.

The main insight afforded by QLC-1 (as described in N2016) lies in its explanation of the resilience of faceted ice growth in the face of inhomogeneities in the overlying vapor field, as a stable limit cycle. Because (as it turns out) that mechanism is the same in the present model, QLC-2, we defer detailed discussion of it to a later section.

QLC-1 also has significant limitations. A structural limitation is that it imposes a fixed time scale for the interconversion of quasi-liquid and ice relative to the other two processes (deposition/sublimation and quasi-liquid/ice equilibration shown in Fig. 1). In real crystals, this time scale could vary from facet to facet, or as a function of temperature and vapor pressure, and should therefore be an adjustable feature of the model. A second shortcoming is that, as presented in N2016, inhomogeneities in the overlying vapor field used in simulations were limited to growth conditions; sublimation conditions were not considered. Moreover, the mathematical representa-

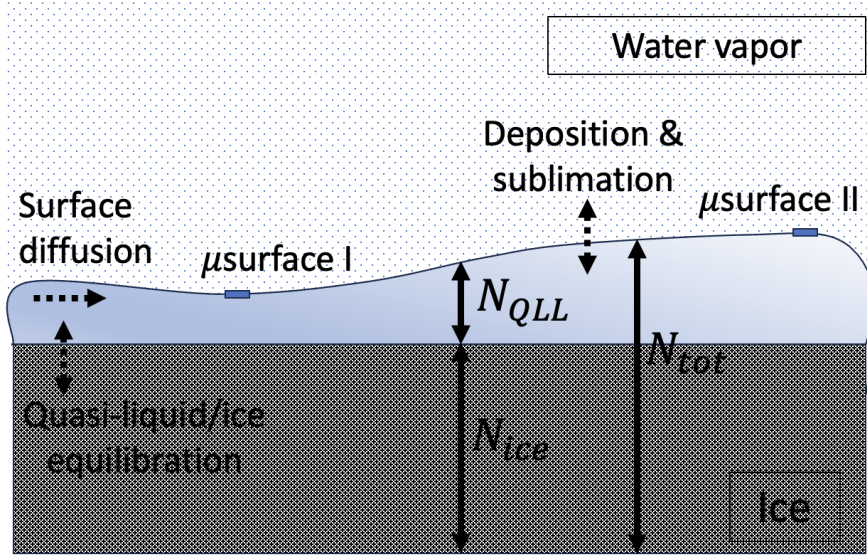


FIG. 1. Visual representation of mesoscale variables N_{tot} , N_{ice} , and N_{QLL} , and processes affecting them, in the QLC-1 and QLC-2 models.

tion of those inhomogeneities in N2016 was speculative: experimental observations of dynamically growing or ablating facets surfaces were not available at that time.

Our goal in the present communication is to redress some of these shortcomings. We do so by formulating a new pair of equations of motion, QLC-2, which differ from QLC-1 in that the time scale of quasi-liquid/ice equilibration dynamics is made explicit, specified by a parameter (t_{eq}). We also revise the representation of the overlying vapor field used in N2016, to provide a closer match to experimental observations that have become available since that time. We show how QLC-1-derived insights into facet stability under growth conditions generalize in QLC-2 to sublimation conditions as well: that is, QLC-2 predicts the phenomenon of *faceted sublimation*. Finally, we formulate a non-dimensional form of the equations of motion, which reveals that a key parameter governing properties of the limit-cycle solutions is the product of the surface diffusion coefficient (D_{QLL}) and the above-mentioned timescale of quasi-liquid/ice equilibration dynamics (t_{eq}). These are described in the next sections.

2. Vapor Field (VF) theory. A key feature of the revised theory presented here is that the water vapor supersaturation profile is obtained from a simulation of the water vapor partial pressure, $P_{vap}(x, y)$, within a space surrounding an idealized, two-dimensional crystal. This is accomplished by integrating the two-dimensional diffusion equation

$$(2.1) \quad \frac{\partial P_{vap}}{\partial t} = D_{vap} \nabla^2 P_{vap}$$

where D_{vap} is the diffusion coefficient of water vapor through air, computed by

$$(2.2) \quad D_{vap} = D_{vap}^o \left(\frac{P_{total}}{1 \text{ atm}} \right) \left(\frac{T}{273 \text{ K}} \right)^{m_D}$$

where we have written P_{total} to indicate the combined pressure of water vapor and the ambient air (see [12]). Values of constants appearing in Eq. 2.2 are given in Table

1.

Eq. 2.1 is solved subject to Neumann and Dirichlet boundary conditions. The latter represents a constant, far-field vapor concentration, $\sigma_{I,\infty}$, at the outside boundary of a two-dimensional simulation space. Neumann conditions represent changes in water vapor at the crystal surface due to crystal growth or sublimation,

$$(2.3) \quad \frac{\partial P_{vap}}{\partial t} = -g_{ice} \times \rho_{ice} \times \frac{RT}{M_{H_2O}}$$

where g_{ice} is the growth rate of the ice surface, ρ_{ice} is the mass density of ice, and M_{H_2O} is the molar mass of water.

The primary goal of conducting VF simulations is to obtain supersaturation profiles adjacent to the ice surface of a given facet, designated $\sigma_I(x)$ or $\sigma_I(y)$, depending on the spatial orientation of the facet. Here, we will use only the former, hence

$$(2.4) \quad \sigma_I(x) = \frac{P_{vap}(x) - P_{vap}^{eq}(T)}{P_{vap}^{eq}(T)}$$

where $P_{vap}^{eq}(T)$ is computed in the usual way, i.e., by the Clausius-Clapeyron equation corresponding to water vapor in equilibrium with ice.

3. Ice surface structure from scanning electron microscopy experiments. In a previous communication [13], some of the present authors reported a methodology for computing quantitative surface topology of ice from images of ice crystals grown in a scanning electron microscope (SEM). The algorithm treats each pixel of an SEM-imaged ice crystal surface as a “microfacet,” i.e., a plane with a specific angular orientation with respect to a reference plane. A surface is then constructed by iteratively optimizing the match between each microfacet’s boundaries to those of its neighbors, via an adaptation of optimal estimation [14]. In [13], the goal was to characterize facet roughness. Here, by contrast, the goal is to characterize curvature across an otherwise smooth facet, on the assumption that such curvature indicates variations in the density of water vapor just above the ice surface. Examples of resulting images are shown in the Results section below.

4. QLC-2 theory. The dynamics of ice crystal growth in the QLC-2 model are given by

$$(4.1) \quad \frac{\partial N_{tot}}{\partial t} = D_{QLL} \nabla_x^2 N_{QLL} + \nu_{kin} \sigma_m(x, N_{QLL})$$

$$(4.2) \quad \frac{\partial N_{QLL}}{\partial t} = \frac{\partial N_{tot}}{\partial t} - \frac{N_{QLL} - N_{QLL}^{eq}(N_{tot})}{t_{eq}}$$

and have the following properties:

1. The operator $D_{QLL} \nabla_x^2$ in Eq. 4.1 represents surface diffusion of the QLL. It operates on N_{QLL} , rather than N_{tot} , to represent the idea that only the QLL (not the underlying ice) is mobile on time scales considered here. The value of D_{QLL} used here is given in Table 2. (When a temperature-dependent value of is desired, we have obtained D_{QLL} from measurements of the self-diffusion coefficient of supercooled water, using the logarithmic formula reported by Price et al [15].)
2. ν_{kin} is the rate at which vapor-phase water molecules strike the ice surface. It is assumed that these stick with 100% efficiency, and thermalize on a picosecond time scale (i.e., instantaneously within the time scale of QLC-2

simulations) [9]. It can be obtained from the temperature according to the Hertz-Knudsen formula,

$$(4.3) \quad \nu_{kin} = P_{vap}^{eq}(T) \times \left(\frac{M_{H_2O}}{2\pi RT} \right)^{1/2}$$

where $P_{vap}^{eq}(T)$ is the equilibrium vapor pressure of ice (obtained by the usual Clausius-Clapeyron formula, using the triple-point temperature of ice as a reference point), R is the molar Boltzmann constant, and M_{H_2O} is the molar mass of water.

3. The effective surface supersaturation, σ_m in Eq. 4.1, is a measure of the supersaturation “experienced” by a given point on the ice surface. It depends not only on a static, imposed surface supersaturation, $\sigma_I(x)$, but also on the volatility of the microsurface at that point. QLC-2 implements this idea as

$$(4.4) \quad \sigma_m(x, t) = \sigma_I(x) - \left(\frac{N_{QLL}(x, t) - (\bar{N} - N^*)}{2N^*} \right) \sigma^o$$

Since μ surface I (corresponding to the minimum QLL thickness appearing in Fig. 1) has $N_{QLL} = (\bar{N} - N^*)$, this formulation ensures that a point on the surface characterized by μ surface I will experience $\sigma_m(x) = \sigma_I(x)$. Likewise, since μ surface II (corresponding to the maximum QLL thickness appearing in Fig. 1) has $N_{QLL} = (\bar{N} + N^*)$, this formulation ensures that a point on the surface characterized by μ surface II will experience a lower effective supersaturation, $\sigma_m(x) = \sigma_I(x) - \sigma^o$. Points of intermediate thickness will experience an interpolated effective supersaturation. We discuss the form of the imposed supersaturation, $\sigma_I(x)$, below, when vapor field simulation results are presented.

4. The form of the right-most term in Eq. 4.2, $\frac{N_{QLL} - N_{QLL}^{eq}(N_{tot})}{t_{eq}}$, is the primary departure of QLC-2 from QLC-1. The function $N_{QLL}^{eq}(N_{tot})$ appearing in the numerator prescribes the thickness of quasi-liquid when it is in equilibrium with the underlying ice,

$$(4.5) \quad N_{QLL}^{eq}(N_{tot}) = \bar{N} - N^* \sin(2\pi N_{tot}).$$

This formulation ensures that the *equilibrium* QLL thickness varies continuously and periodically from the thin μ surface I (as just mentioned, having thickness $\bar{N} - N^*$) to the thick μ surface II (thickness $\bar{N} + N^*$). The denominator, t_{eq} , is a first-order relaxation constant describing the time scale of quasi-liquid/ice equilibration, shown schematically at the left-hand side of Fig. 1.

4.1. Nondimensionalization. We nondimensionalize Eqs.4.1 and 4.2 through the following transformations:

$$(4.6) \quad \tau = \frac{t}{t_{eq}}$$

and

$$(4.7) \quad \chi = \frac{x}{(D_{QLL} t_{eq})^{\frac{1}{2}}},$$

resulting in the nondimensional forms

$$(4.8) \quad \frac{\partial N_{tot}}{\partial \tau} = \nabla_{\chi}^2 N_{QLL} + \omega_{kin} \sigma_m(\chi, N_{QLL})$$

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$$(4.9) \quad \frac{\partial N_{QLL}}{\partial \tau} = \frac{\partial N_{tot}}{\partial \tau} - \left(N_{QLL} - N_{QLL}^{eq}(N_{tot}) \right)$$

where $\omega_{kin} = t_{eq} \nu_{kin}$. This transformation exposes a key feature of the QLC-2 model, namely, that D_{QLL} appears only as the product $D_{QLL} \times t_{eq}$ in the rescaled spatial coordinate χ .

The above equations are written assuming a one-dimensional context (hence expressions like $\sigma_m(x, N_{QLL})$). It is sometimes useful, however, to reduce that dimensionality to zero, by setting $D_{QLL} = 0$. Trajectories of the zero-dimensional system are discussed briefly below.

5. Methods. The vapor pressure equation of motion (Eq. 2.1) is integrated using Euler’s method, with parameters given in Table 1. The QLC-2 equations of motion are solved using the LSODA method, which switches dynamically between between the nonstiff Adams method and the stiff BDF method.

6. Results.

6.1. Bergian and anti-Bergian patterns of vapor concentrations. Selected vaporfield results are shown in Fig. 2. The upper panel of column (a) of this figure demonstrates what we define here to be a “Bergian pattern,” namely, where the supersaturation immediately above a growing faceted surface is higher at facet corners than it is at the facet center. The lower panel of column (a) shows the supersaturation immediately above the crystal surface, along with a parabolic best-fit representation,

$$(6.1) \quad \sigma_I(x) \approx \sigma_{I,\text{corner}} \times \left(c_r \left(\frac{x}{L} \right)^2 + (1 - c_r) \right)$$

where x specifies position on the ice surface, and c_r (for “center reduction”) is the fractional reduction of supersaturation at facet centers relative to facet corners,

$$(6.2) \quad c_r = 1 - \frac{\sigma_{I,\text{middle}}}{\sigma_{I,\text{corner}}}$$

Column (b) exhibits an “anti-Bergian pattern,” where σ_I immediately above a growing faceted surface is *lower* at facet corners than it is at the facet center. This is due to the adjacent facets growing so fast that they deplete nearby air of water vapor. Column (c) returns to a “Bergian pattern,” but in the context of sublimation. That is, σ_I immediately above a sublimating faceted surface is more extreme (more negative) at facet corners relative to the facet center.

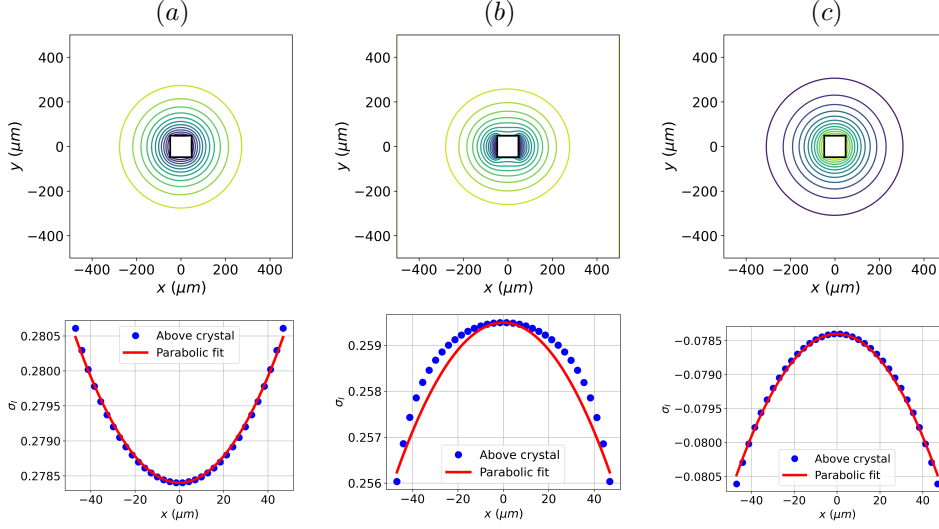


FIG. 2. (a) Steady-state water vapor supersaturation around a square ice crystal indicated by black lines at the center, obtained by solving Eq. 2.1, subjected to Dirichlet supersaturated conditions in the far field and negative Neumann flux conditions at the crystal surface (Eq. 2.3), representing equal growth rates at all four surfaces of the crystal. In the upper panel, contours of increasing diameter correspond to higher supersaturation values. The lower panel shows supersaturations that would be felt by a $\mu\text{surface I}$ microfacet at the crystal surface due to the vapor pressure immediately adjacent to it, obtained using Eq. 2.4. (b) Similar to (a), but with Neumann conditions such that growth rates at the left and right sides of the crystal are four times greater than at the upper and lower facets. (c) Similar to (a), but with subsaturated far-field water vapor and positive Neumann fluxes at the surface, corresponding to sublimation at the crystal surface. In the upper panel, contours of increasing diameter correspond to more extreme subsaturation.

6.2. Results from scanning electron microscopy. Figure 3 shows the results of retrieving surface heights using SEM/GNBF on a prismatic facet of a growing ice crystal. The distinct concave pattern in the crystallographic a-axis direction is consistent with the anti-Bergian pattern of water vapor concentration for a growing ice crystal shown in Panel (b) of Fig. 2. The two prismatic facets adjacent to the prismatic facet being imaged are expected to be faster-growing because of their closer proximity to the copper platform, whose temperature is set to be cold when crystal growth within the SEM chamber is desired.

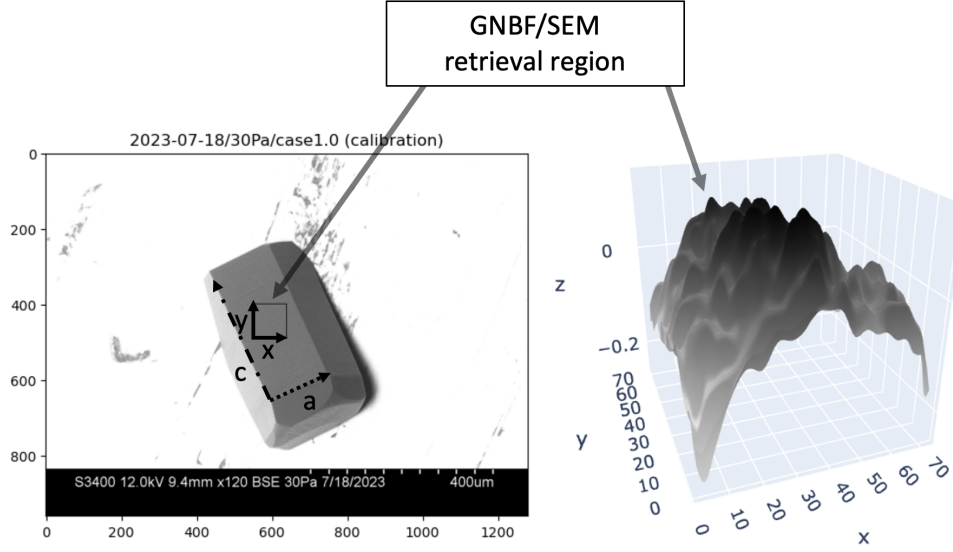


FIG. 3. Scanning electron image results of a growing ice crystal highlighting the region analyzed with SEM/GNBF (left), and the results of SEM/GNBF surface reconstruction (right). Crystallographic axes “a” and “c” are indicated as dashed and dashed-dot lines, respectively.

6.3. QLC-2 continuum model results (0-dimensional).

Figure 4 shows results from three runs of the zero-dimensional (i.e., $D_{QLL} = 0$) version of QLC-2. Three different initial conditions, depicted as red-colored circles, all quickly converge to a limit cycle that appears stable and whose thickness is periodic in time. As the quasiliquid layer grows thicker, the conversion of quasiliquid to crystalline ice outpaces the rate of deposition and the quasi-liquid layer shrinks. When the quasiliquid is thin, the rate of deposition outpaces the conversion to ice, and the quasiliquid layer grows. The result is a periodic variation in the QLL thickness over time, even as the crystal as a whole grows.

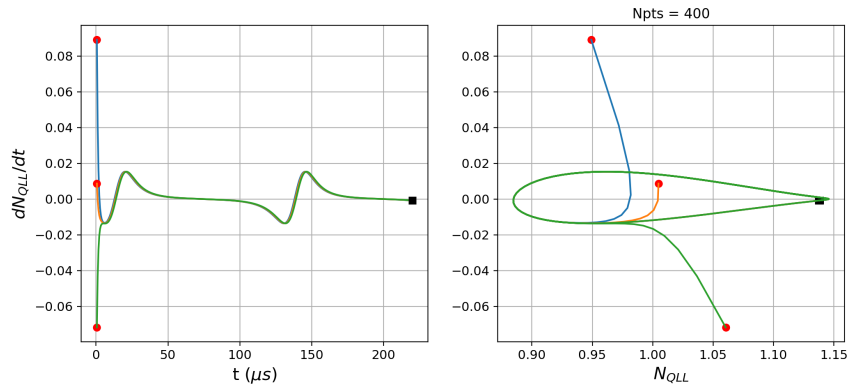


FIG. 4. Three trajectories of a zero-dimensional version of QLC-2 showing convergence to the stable limit cycle.

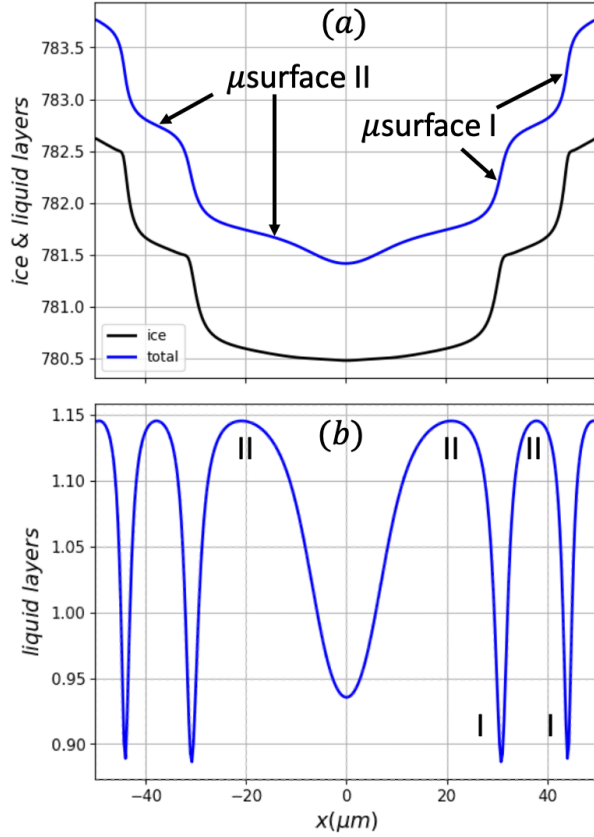


FIG. 5. An ice surface covered by QLL, as simulated by QLC-2. Panel (a) shows total ice thickness and the underlying crystalline ice thickness. Panel (b) shows the difference between these, i.e., the thickness of the quasiliquid layer.

6.4. QLC-2 continuum model results (1-dimensional). A snapshot from a typical trajectory of the one-dimensional version of QLC-2 is shown in Fig. 5. Panel (a) shows that the surface is structured like a staircase, with broad, flat parts (“treads”), separated by abrupt changes in the surface height (“risers”). Panel (b) shows that treads have QLL thickness corresponding to the thick $\mu_{\text{surface II}}$, while risers have thickness close to the thinner $\mu_{\text{surface I}}$. Dynamically, this means that risers experience a higher effective supersaturation (σ_m) than treads, for the same imposed supersaturation (σ_I), and therefore can be expected to grow faster. As a consequence, over time, the staircase appears to move toward the facet center, while new steps form at facet corners. Animations in the Supplemental Information display these properties dynamically.

We can use Figure 5 to describe, qualitatively, a mechanism for uniform growth. The first part of the mechanism relies on the notion of “diffusive slowdown”: diffusion of the QLL slows the average rate of growth of the surface. The reason for diffusive slowdown is that risers are constantly receiving an influx of quasiliquid from nearby treads, because the latter have thicker quasiliquid. The constant loss of $\mu_{\text{surface-I}}$ -like coverage slows down growth overall.

The second part of the mechanism relies on the notion that diffusive slowdown is not the same across a given surface: there is more diffusive slowdown where the staircase is steeper, and therefore the steps are closer together. Panel (b) of Fig. 5 shows that transitions in quasiliquid thickness around riser positions become ever sharper as one proceeds from facet center to facet corners. That sharpness leads to faster diffusion into riser positions, hence greater diffusive slowdown, as one nears facet corners.

The third part of the mechanism of uniform growth concerns the way greater diffusive slowdown at facet corners acts to compensate for the faster potential for growth there due to higher σ_I (assuming a Bergian pattern of the overlying water vapor concentration). It turns out that – because of nonlinearities in the equations of motion – surfaces are spontaneously attracted to a state in which this compensation is *exact* and *stable*. More precisely, the system evolves spontaneously toward a stable limit cycle for N_{QLL} , and a periodic traveling wave pattern for N_{tot} . The result is the uniform growth rate across the entire surface associated with faceted growth.

A quantitative version of the above three-part mechanism for uniform (faceted) growth in the presence of inhomogeneities in the overlying vapor field was provided by N2016. N2016 did not, however, probe the full extent of the mechanism, as we elaborate next with the help of Fig. 6. In that figure, we see a modeled ice crystal surface subjected to three overlying vapor field profiles. Each simulation begins with a flat surface, and each leads to a distinctive limit-cycle/periodic traveling wave solution for N_{QLL} , and N_{tot} , respectively. Close examination of those solutions provides a number of insights:

- Case (a) shows the result of imposing a convex (Bergian) supersaturation profile over an initially flat surface. After ≈ 150 ms, the surface has converged to a convex, staircase-shaped limit cycle that mimics the curvature of the overlying supersaturation profile. The difference between the maximum N_{tot} and the minimum N_{tot} in this staircase defines “ N_{steps} ”, plotted as function of time in the right-hand panel. This panel shows that a limit cycle has been achieved. (See the Supplemental Information for an animation.)
- Case (b) imposes a sinusoidal pattern of supersaturation. We do not expect such patterns to occur in real crystals, the intent is rather to probe whether the domain of the attractor spans such inhomogeneities. The answer is yes: after a time interval of only ≈ 60 ms, in fact, an initially-flat ice surface exposed to this vapor profile has converged to a limit cycle. Moreover, once again, the structure of the ice limit-cycle surface mimics that of the overlying supersaturation profile.
- Case (c) imposes a subsaturated vapor profile, concave-shaped, and therefore (like (a)) Bergian. After ≈ 500 ms, an initially-flat ice surface has also converged to a limit cycle that mimics the curvature of the overlying vapor profile, with a staircase-like structure. Unlike cases (a) and (b), this is an *sublimating* limit cycle. Careful inspection, moreover, reveals a surprise: here (and in all other sublimating limit cycles we have found), *treads are thin and μ surface-I-like*, while *risers are thick and μ surface-II-like*. The rationale for this begins with the observation that in a subsaturated environment, the least-volatile microsurface (μ surface I) is the most resistant to sublimation.

The take-home message of Fig. 6 is that QLC-2 supports a set of limit-cycle attractors that span a greater range of overlying vapor field concentrations than pre-

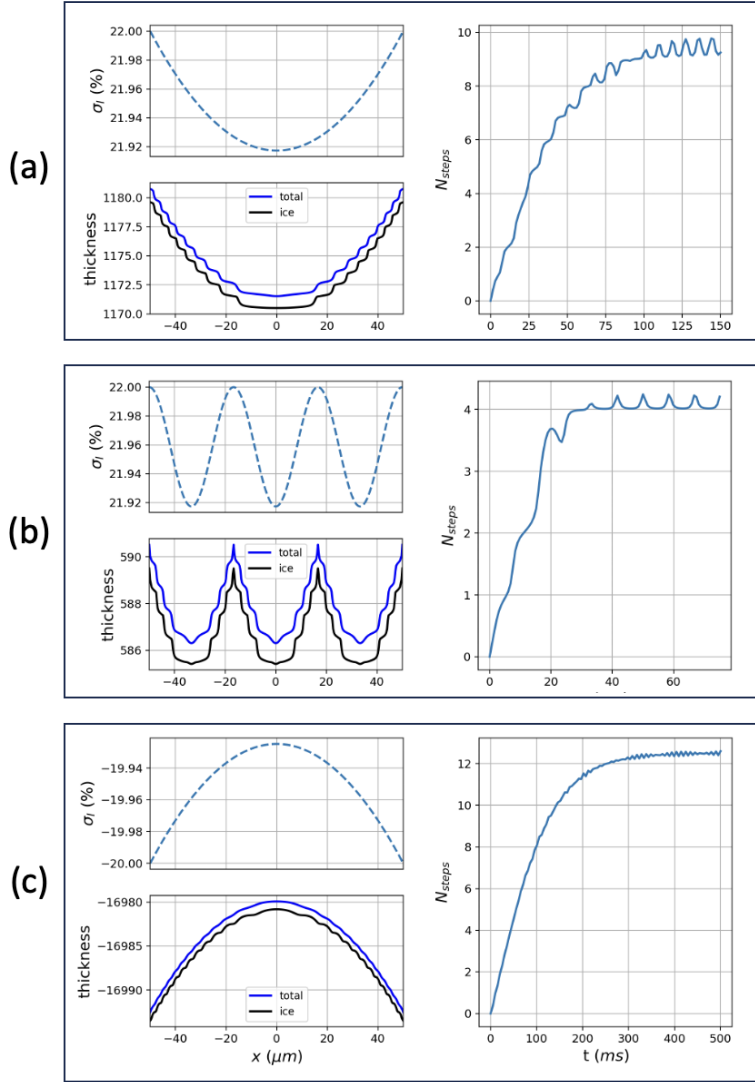


FIG. 6. Dynamics of surfaces subjected to various overlying water vapor concentrations. (a) The upper left displays the parabolic-shaped supersaturation imposed across the ice surface. The panel below that shows thicknesses (total ice, and ice only) of the converged limit cycle. The difference between the maximum N_{tot} and the minimum N_{tot} defines “ N_{steps} ”, which is plotted as function of time in right-hand panel. The leveling-off of “ N_{steps} ” after about 150 ms simulated time shows that a limit cycle has been reached. (b) Dynamics when a corrugated (sinusoidal) supersaturation pattern is imposed. (c) Dynamics under parabolic Bergian subsaturation conditions.

viously understood (in N2016). Specifically, these attractors include both Bergian (V-shaped) and anti-Bergian (Λ -shaped vapor supersaturation patterns, as well as subsaturated conditions. The latter leads to the prediction of *faceted sublimation*, in which μ surfaces associated with treads and steps are swapped relative to the associations during growth conditions.

Figure 7 explores $N_{\text{steps}}^{\text{limit}}$, the leveled-off maximum minus minimum values of N_{tot} such as appearing in the right-hand column of Fig. 6, as a function of parameters c_r

323 and ω_{kin} .

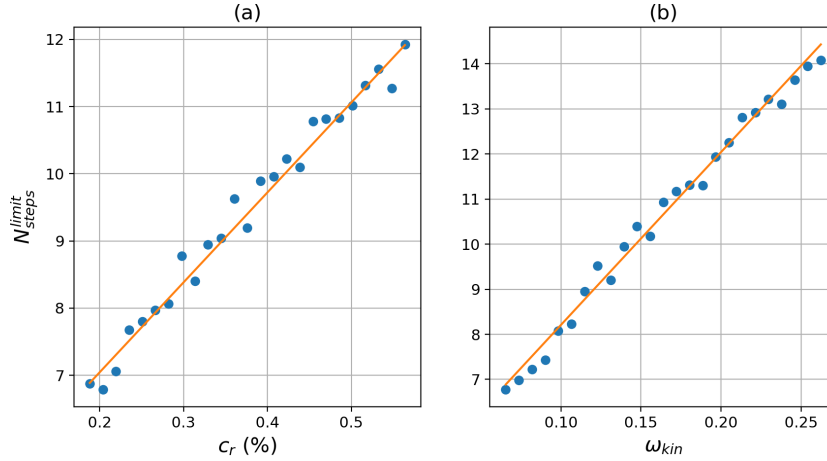


FIG. 7. N_{steps}^{limit} as a function of the center reduction (c_r) and the non-dimensional kinetic velocity (ω_{kin}). Lines indicate least-squared fits to Eq. 6.3.

324 The linearity and commonality of the y-intercept in these figures suggests the
325 bilinear form

$$326 \quad (6.3) \quad N_{steps}^{limit} \approx m c_r \omega_{kin} + b$$

327 The solid lines in the figure use Eq. 6.3 with $m = 102 \text{ percent}^{-1}$ (if c_r is expressed
328 in percent) and $b = 4.4$ layers.

329 **7. Summary and Discussion.** We have found that faceting exhibited by solu-
330 tions of the QLC-2 model is highly resilient to inhomogeneity in the overlying vapor
331 field. This is demonstrated by the trajectories shown in Fig. 6, which represent
332 only a few cases out of many that we have examined: in fact, every smoothly vary-
333 ing overlying vapor field that we have examined leads to a distinctive and stable
334 limit-cycle/periodic traveling wave solution for N_{QLL} , and N_{tot} , respectively. The
335 mechanism enabling this resilience, “diffusive slowdown,” consists of the tendency of
336 any part of a facet surface subjected to higher overlying vapor concentration to recon-
337 struct itself in such a way that it becomes less efficient at capturing and converting
338 vapor into ice.

339 The QLC-2 model also predicts that growing facets will develop mesoscale profiles
340 that are convex or concave, depending on whether the inhomogeneity of the overlying
341 vapor field is Bergian (i.e., convex-shaped vapor supersaturation profiles) or anti-
342 Bergian (concave-shaped vapor supersaturation profiles). Similar patterns are found
343 in reconstructions of SEM-grown ice crystals, such as the one shown in Fig. 3. (We
344 hasten to note, however, that the vertical relief of the patterns that we are able to
345 generate numerically using QLC-2 are of just a few dozen molecular layers, as opposed
346 vertical relief observed in SEM experiments, which can be thousands of molecular
347 layers.)

348 It is of interest to compare the present results to the familiar phenomenon of
349 “Turing patterns.” In Turing’s formal treatment [16], spatial patterns originating

from reaction-diffusion dynamics are predicted to exhibit a characteristic distance – for example, a characteristic separation between stripes on a zebra’s skin. That distance, in Turing’s formulation, depends on the square root of the surface diffusion coefficient of key reactants. Here, by contrast, the spatial transformation leading to the nondimensional form of QLC-2, specified by Eq. 4.7, leads almost trivially to an analogous conclusion, namely, that structural features that appear on the surface of ice will scale in proportion to the square root of D_{QLL} , the surface diffusion coefficient of quasiliquid. Moreover, this conclusion is not limited in the same way that Turing’s theory is, in that Turing’s conclusions proceed from a sensitivity analysis to perturbations to an initially homogeneous distribution of chemical species, whereas the transformation embodied in Eq. 4.7 comes with no such caveat: it applies throughout a simulation in QLC-2. The work of reference [17] may also be relevant here, in the dependence of the stability of traveling pulse solutions of the FitzHugh–Nagumo equation on time scales intrinsic to that model.

A meteorological implication of the above concerns the morphology of cirrus ice crystals as they fall through Earth’s atmosphere. As such crystals fall, they will encounter increased pressure, hence smaller D_{vap} , but also increased temperatures (unless there is an atmospheric inversion), hence larger D_{vap} . If the effect of increased pressure dominates, QLC-2 predicts larger c_r , and therefore enhanced growth at facet corners and a greater propensity toward hollowed crystal structures. Although there is no single observational datum that would definitively resolve these predictions, we can comment that in exceptionally cold regions (such as the Antarctic Plateau), or even in mid-latitudes where high-altitude cryo-capture of ice crystals on ground-launched balloons is possible, observations have shown that cirrus cloud ice crystals are frequently hollowed [18, 19, 20].

We should note that the results presented here do not consider variations in parameters σ^o , N^* , and $\bar{N}$, because those values are highly uncertain. Because different facet types (basal, prismatic, or pyramidal) have distinct underlying crystal cell structures, it is to be expected that their quasi-liquid properties (σ^o , N^* , and $\bar{N}$) will differ from those presented here. More work will be necessary to learn whether those differences translate into any qualitatively new phenomena.

Finally, although the context here is vapor-exposed water ice, we speculate that something akin to QLC-2 theory could be adapted to more general contexts (e.g., to crystals grown from solutions, or to vapor-grown crystals other than water). Firstly, the expression for the kinetic velocity ν_{kin} , Eq. 4.3, is easily modified for volatile substances other than water vapor, and one can imagine plausible analogs for solutes as well. Secondly, volatilities of the two limiting-case microsurfaces (μ surfaces I and II) assumed here for the purpose of computing effective supersaturation (σ_m in Eq. 4.2) are more generally expressible in terms chemical potentials, which would apply to solutes as well. Thirdly, it has long been recognized that surface diffusion plays an essential role in crystal growth from solutions as well as from vapor [21, 22], and that a mobile surface layer “which constitutes a transition” between the solid crystal and the medium through which molecules move to add to the crystal [7] is key to that process. Two critical unknowns, in our view, are whether two limiting-case microsurfaces of a mobile surface layer in fact exist, and if so, then what their equilibrium thicknesses (analogous to Eq. 4.5) and chemical potentials may be.

TABLE 1
Default vapor field parameters.

Description	Value
Simulation space dimensions	$X = Y = \pm 500 \mu\text{m}$
Time step for integration	$dt = 5 \times 10^{-5} \mu\text{s}$
Time interval for integration	$t_{max} = 1 \mu\text{s}$
Spatial discretization	$dx = dy = 2.8 \mu\text{m}$
Diffusion coefficient at $T = 273 \text{ K}$, $P_{total} = 1 \text{ atm}$	$D_{vap}^o = 21.9 \mu\text{m}^2/\mu\text{s}$
Temperature	$T = 240 \text{ K}$
Total pressure	$P_{total} = 50 \text{ Pa}$
Diffusion Temperature-correction exponent	$m_D = 1.86$
Diffusion coefficient under ambient conditions	$D_{vap} = 3.49 \times 10^4 \mu\text{m}^2/\mu\text{s}$
Far-field water vapor supersaturation	$\sigma_I = 0.25$
Mass density of ice	$d_{ice} = 0.9 \text{ g/cm}^2$
Growth rate of ice surface	$g_{ice,s} = 2 \mu\text{m/s}$

TABLE 2
Default QLC-2 Parameters.

Description	Value
Facet width ($\pm L$)	$L = 50 \mu\text{m}$
Spatial discretization	$dx = 0.3 \mu\text{m}$
Surface diffusion coefficient	$D_{QLL}^o = 0.0005 \mu\text{m}^2/\mu\text{s}$
Volatility difference between μ surfaces I and II	$\sigma^o = 0.2$
Mean thickness of QLL	$\bar{N} = 1 \text{ layer}$
Variability in QLL thickness	$N^* = 0.9/2\pi$
Temperature	$T = 240 \text{ K}$
Kinetic velocity	$\nu_{kin} = 40 \mu\text{m/s}$
Nondimensional kinetic velocity	$\omega_{kin} = 0.13$
Center reduction	$c_r = 0.376\%$

8. Appendix.

REFERENCES

- [1] Ukichiro Nakaya. *Snow crystals: natural and artificial*. Harvard University Press, 1954.
- [2] Kenneth G Libbrecht. The formation of snow crystals: subtle molecular processes govern the growth of a remarkable variety of elaborate ice structures. *American Scientist*, 95(1):52–59, 2007. Publisher: JSTOR.
- [3] Emma Järvinen, Bastiaan Van Dienenhoven, Nathan Magee, Steven Neshyba, Martin Schnaiter, Guanglang Xu, Olivier Jourdan, David Delene, Fritz Waitz, Simone Lolli, and others. Ice crystal complexity and link to the cirrus cloud radiative effect. *Clouds and Their Climatic Impacts: Radiation, Circulation, and Precipitation*, pages 47–85, 2023.
- [4] Pablo Llombart, Eva G Noya, and Luis G MacDowell. Surface phase transitions and crystal habits of ice in the atmosphere. *Science advances*, 6(21):eaay9322, 2020.
- [5] Jerry Y Harrington and Gwenore F Pokrifka. Approximate models for lateral growth on ice crystal surfaces during vapor depositional growth. *Journal of the Atmospheric Sciences*, 78(3):967–981, 2021.
- [6] T Kuroda and R Lacmann. Growth kinetics of ice from the vapour phase and its growth forms. *Journal of Crystal Growth*, 56(1):189–205, 1982.
- [7] WF Berg. Crystal growth from solutions. *Proceedings of the Royal Society of London. Series A-Mathematical and Physical Sciences*, 164(916):79–95, 1938. Publisher: The Royal Society London.

- [8] C Jayaprakash, WF Saam, and S Teitel. Roughening and facet formation in crystals. *Physical review letters*, 50(25):2017, 1983.
- [9] Steven Neshyba, Erin Nugent, Martina Roeselová, and Pavel Jungwirth. Molecular Dynamics Study of Ice- Vapor Interactions via the Quasi-Liquid Layer. *The Journal of Physical Chemistry C*, 113(11):4597–4604, 2009.
- [10] Ken-ichiro Murata, Ken Nagashima, and Gen Sazaki. How do ice crystals grow inside quasiliquid layers? *Physical Review Letters*, 122(2):026102, 2019.
- [11] Steven Neshyba, Jonathan Adams, Kelsey Reed, Penny M Rowe, and Ivan Gladich. A quasiliquid mediated continuum model of faceted ice dynamics. *Journal of Geophysical Research: Atmospheres*, 121(23):14–035, 2016.
- [12] The Engineering ToolBox (2018). Air - Diffusion Coefficients of Gases in Excess of Air. [<https://www.engineeringtoolbox.com/air-diffusion-coefficient-gas-mixture-temperature-d.2010.html>, accessed 4/23/2025].
- [13] Nicholas Butterfield, Penny M Rowe, Emily Stewart, David Roesel, and Steven Neshyba. Quantitative three-dimensional ice roughness from scanning electron microscopy. *Journal of Geophysical Research: Atmospheres*, 122(5):3023–3041, 2017.
- [14] Clive D Rodgers. *Inverse methods for atmospheric sounding: theory and practice*, volume 2. World scientific, 2000.
- [15] William S Price, Hiroyuki Ide, and Yoji Arata. Self-diffusion of supercooled water to 238 K using PGSE NMR diffusion measurements. *The Journal of Physical Chemistry A*, 103(4):448–450, 1999. Publisher: ACS Publications.
- [16] AM Turing. The Chemical Basis of Morphogenesis: Philosophical Transactions of the Royal Society of London. Ser. B. In *Biol. Sci*, volume 237, 1952.
- [17] Gianni Arioli and Hans Koch. Existence and stability of traveling pulse solutions of the FitzHugh–Nagumo equation. *Nonlinear Analysis: Theory, Methods & Applications*, 113:51–70, 2015.
- [18] NB Magee, Andrew Miller, M Amaral, and A Cumiskey. Mesoscopic surface roughness of ice crystals pervasive across a wide range of ice crystal conditions. *Atmospheric Chemistry and Physics*, 14(22):12357–12371, 2014.
- [19] Nathan Magee, Katie Boaggio, Samantha Staskiewicz, Aaron Lynn, Xuanyi Zhao, Nicholas Tusay, Terance Schuh, Manisha Bandamede, Lucas Bancroft, David Connolly, and others. Captured cirrus ice particles in high definition. *Atmospheric Chemistry and Physics Discussions*, 2020:1–27, 2020.
- [20] Von P Walden, Stephen G Warren, and Elizabeth Tuttle. Atmospheric ice crystals over the Antarctic Plateau in winter. *Journal of Applied Meteorology*, 42(10):1391–1405, 2003.
- [21] P. Bennema. The importance of surface diffusion for crystal growth from solution. *Journal of Crystal Growth*, 5(1):29–43, 1969.
- [22] W-K. Burton, N t Cabrera, and FC Frank. The growth of crystals and the equilibrium structure of their surfaces. *Philosophical Transactions of the Royal Society of London. Series A, Mathematical and Physical Sciences*, 243(866):299–358, 1951. Publisher: The Royal Society London.